

USER ACCEPTANCE TESTING

UAT Execution & Report Submission

EduGenie: Google Gemini Powered Learning Assistant

Report Date	02 October 2026
Team ID	SWTID -2026-1665
Project	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

UAT status: Prepared from supplied project documentation; formal UAT pass/fail evidence was not included.

1. Purpose of Document

This User Acceptance Testing (UAT) report documents the intended acceptance-test coverage for EduGenie, an AI-powered educational assistant built with a FastAPI backend and an HTML/CSS frontend. The report translates the project's documented user functions and functional-testing scope into a structured UAT format.

Important evidence note: The supplied project documentation describes the implemented features and lists functional-testing activities, but it does not contain a completed UAT execution log, tester observations, screenshots for each test, or pass/fail results. Status values are therefore shown as Not recorded.

2. UAT Scope & Product Coverage

Area	Documented Coverage	Expected User Outcome
Q&A	Gemini-powered question answering through /qa	Concise answer to a user question
Explanation	LaMini-Flan-T5 explanation module through /explain	Complex concept simplified for learners
Quiz	Gemini quiz generation through /quiz	3 MCQs, 4 options each, with correct answer
Summary	Gemini summarization through /summarize	Concise, easy-to-understand summary
Learning Path	Gemini recommendations through /learn/recommendations	Beginner-to-advanced learning path with resources
Web Interface	Task dropdown, text area, submit button and responsive result area	Simple interaction and real-time result display

3. Test Case Analysis

ID	Functional Area	Test Input / Action	Expected Result	Status
UAT-01	Question & Answer	Enter an academic/general question.	System returns a concise AI-generated answer.	Not recorded
UAT-02	Concept Explanation	Select Explain and enter a topic.	System provides a simplified, readable explanation using the explanation module.	Not recorded
UAT-03	Quiz Generation	Provide a topic/passage and select Quiz.	System generates 3 MCQs with 4 options each and a correct answer.	Not recorded
UAT-04	Summarization	Enter a long educational passage and select Summary.	System returns a concise summary retaining key information.	Not recorded

UAT-05	Learning Recommendations	Enter a learning topic and select Recommend Path.	System provides a structured beginner-toadvanced learning path with resources.	Not recorded
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4. Defect Analysis

The original template contains sample defect counts. Those figures are not attributed to EduGenie in the supplied project documentation and have not been reused as project results.

Resolution	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Subtotal	Evidence
By Design	0	0	0	0	0	No UAT defect data supplied
Duplicate	0	0	0	0	0	No UAT defect data supplied
External	0	0	0	0	0	No UAT defect data supplied
Fixed	0	0	0	0	0	No UAT defect data supplied
Not Reproduced	0	0	0	0	0	No UAT defect data supplied
Skipped	0	0	0	0	0	No UAT defect data supplied
Won't Fix	0	0	0	0	0	No UAT defect data supplied
TOTAL	0	0	0	0	0	Execution defect log not provided

5. Technical & Deployment Context

The project documentation identifies Python 3.10+, FastAPI, HTML/CSS, Google Gemini API, Uvicorn and Jinja2 as key prerequisites/components. The backend exposes RESTful endpoints for the major learning functions. The application is run locally with uvicorn main:app --reload.

Component	Documented Role
FastAPI	Backend framework and REST API layer
Gemini 1.5 Pro	Q&A, summarization, quiz generation and learning paths
LaMini-Flan-T5-783M	Local concept explanation
HTML + CSS	Web interface and responsive presentation
Uvicorn	ASGI server used to run the application locally
Jinja2	HTML templating engine

Documented local deployment: <http://127.0.0.1:8000/>

6. Open Issues / UAT Considerations

The project documentation identifies challenges including AI-response accuracy, performance across devices, personalization versus general usability, prompt fine-tuning, UI clarity and data-privacy concerns. The supplied material does not classify these as open UAT defects.

Consideration	Documented Context	UAT Treatment
AI response accuracy	Accuracy of generated responses was identified as a development challenge.	Validate with representative learner questions during UAT.
Performance	Performance across devices was identified as a challenge.	Record response behavior across intended test devices.
Personalization / usability	Balancing personalization with general usability was identified as a challenge.	Capture learner feedback on clarity and usefulness.
Data privacy	Data privacy concerns were identified during development.	Confirm applicable project privacy requirements before release.

7. UAT Summary & Sign-Off

Based on the supplied project documentation, EduGenie has documented implementations for the five core educational functions and a web interface that connects user input to the FastAPI backend. However, a completed UAT execution record with tester names, execution dates, evidence and pass/fail outcomes was not included.

Product under UAT	EduGenie: Google Gemini Powered Learning Assistant
UAT scope	Q&A, explanation, quiz, summarization, learning recommendations and web interaction
Documented test execution result	Not recorded in supplied documentation
Defect log	Not supplied; no project defect counts asserted in this report
Release/UAT decision	Requires formal execution and stakeholder sign-off

Prepared from: EduGenie project documentation supplied with this request and the supplied UAT Report Template.

Note: 127.0.0.1 is a loopback address and is normally accessible only from the machine running the application; it is recorded here as the documented deployment address, not as a public release URL.